

Grade 8
Unit 7
Lesson 9 Practice

Name Answer key
Date _____
Period _____

In 1960 the average population of Florida was 4.95 million. Then in 2020, the average population had increased to 21.73 million. Assume that the pattern was linear.

1. Define the quantities in this relationship using variables.

$x = \text{time (years)}$
 $y = \text{population (in millions)}$

2. Find the slope and y-intercept for this relationship.

$$\frac{21.73 - 4.95}{2020 - 1960} = \frac{16.78}{60} \approx 0.28 = m \quad 1960 = \text{year } 0 \rightarrow (0, 4.95) = b$$

3. Write an equation to represent the relationship between the average population and the number of years since 1960.

$$y = 0.28x + 4.95$$

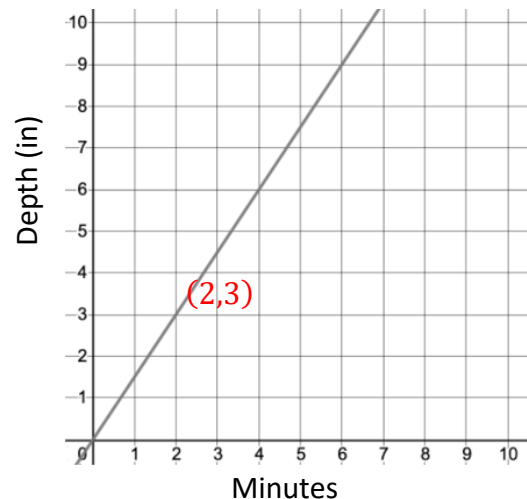
George is filling his aquarium with water. The graph shows how the depth, in inches, of the water changes over time, in minutes.

4. Define the variables you will use.

$x = \text{time (minutes)}$
 $y = \text{depth (in.)}$

5. Find the slope and y-intercept of the graph.

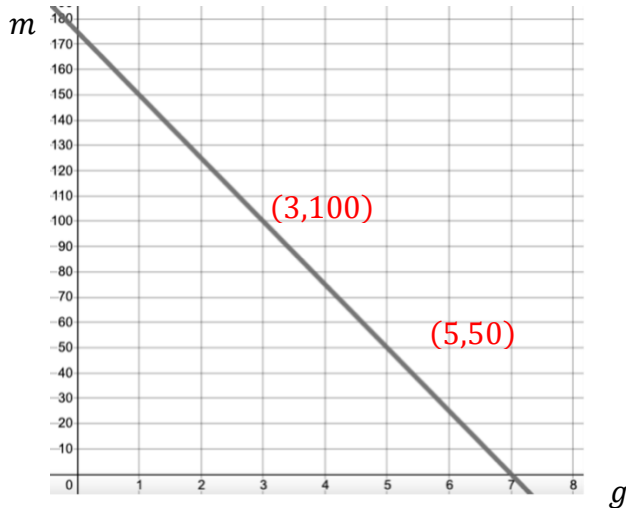
$$\frac{3 - 0}{2 - 0} = \frac{3}{2} = m \quad (0, 0) = b$$



6. Write a linear equation in slope-intercept form that represents this relationship.

$$y = \frac{3}{2}x + 0 \quad \text{or} \quad y = \frac{3}{2}x$$

For each linear relationship, define the variables and then write the equation of the linear relationship using the defined variables.

Linear Relationship	Variables	Equation										
7. Daniel is riding his four-wheeler. His gas usage is shown in the graph. Let m represent the number of possible miles and g represent the number of gallons used. 	$x = g =$ gallons of gas $y = m =$ miles	$m = \frac{50-100}{5-3} = \frac{-50}{2} = -25$ y-int (0,175) $y = -25x + 175$ $m = -25g + 175$										
8. You are at your friend's house playing video games for 8 hours. Your score is 1600 points.	$x =$ time, in hours $y =$ number of points	$\frac{1600}{8} = 200 = m$ y-int (0,0) $y = 200x$										
9. Since 2010, the price Smith's have put money into their savings account. <table border="1" data-bbox="300 1455 621 1652"><thead><tr><th>Year</th><th>Savings</th></tr></thead><tbody><tr><td>2010</td><td>\$14,300</td></tr><tr><td>2013</td><td>\$21,800</td></tr><tr><td>2019</td><td>\$36,800</td></tr><tr><td>2021</td><td>\$41,800</td></tr></tbody></table>	Year	Savings	2010	\$14,300	2013	\$21,800	2019	\$36,800	2021	\$41,800	$x =$ year $y =$ savings (\$)	$\frac{21800-14300}{13-10} = \frac{7500}{3} = 2500$ $y = 2,500x + 14300$
Year	Savings											
2010	\$14,300											
2013	\$21,800											
2019	\$36,800											
2021	\$41,800											
10. A 9 inch candle is burning at $\frac{1}{2}$ inch per hour.	$x =$ time $y =$ height (in.)	$y = \frac{-1}{2}x + 9$										